

VIT BHOPAL UNIVERSITY

SCHOOL OF COMPUTING SCIENCE & ENGINEERING

EVALUATED PROJECT

FUNDAMENTALS OF AI & ML



ECO-LOGIC

THERMAL INTELLIGENCE

A Machine Learning Approach to Human-Centric
Sustainable HVAC Optimization

PROJECT SUBMISSION DETAILS

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1. Abstract

The **Eco-Logic Thermal Intelligence System** is an interactive, Machine Learning-based application designed to optimize Heating, Ventilation, and Air Conditioning (HVAC) systems. By treating human occupants and active electronic devices as dynamic heat sources, the system utilizes a **Decision Tree Regressor** to predict the most energy-efficient thermostat setting. This project bridges the gap between Digital Literacy and Environmental Sustainability, demonstrating how Artificial Intelligence can be leveraged to reduce carbon footprints while maintaining optimal human comfort.

2. The Problem: Why It Matters



The Core Issue

Traditional HVAC systems in homes, universities, and corporate buildings operate on rigid, static logic. They rely solely on ambient room temperature, adhering to a "Set-and-Forget" methodology. These systems fail to account for internal, dynamic heat variables:

1. **Human Thermal Radiation**: An average human resting in a room emits approximately 100 Watts of heat.
2. **Electronic Heat Dissipation**: Active laptops and desktop computers emit between 50W to 300W of thermal energy.



Why It Matters

When 50 students sit in a computer lab with 50 running PCs, they collectively generate massive amounts of heat. A traditional heater will still blindly consume grid electricity to warm the room to a set point, wasting immense amounts of energy. Conversely, in summer, the AC might struggle to cool the room if it doesn't preemptively account for this internal heat load.

- **Economic Impact**: HVAC accounts for nearly 40% of a commercial building's energy consumption.
- **Environmental Impact**: Wasted electricity directly translates to unnecessary CO_2 emissions.

Solving this problem aligns perfectly with the goals of Digital Literacy: utilizing modern computational tools to solve pressing, real-world ecological challenges.

3. The Approach: Solving the Problem

To build a proactive, intelligent climate control system, I implemented a Supervised Machine Learning approach using **Python** and **Scikit-Learn**.



Data Simulation & Feature Engineering

Since access to thousands of real-world, labeled IoT thermal records is restricted, I engineered a high-fidelity synthetic dataset consisting of 600 unique environmental scenarios.

- **Inputs (Features)**: Outside_Temp_C (-45°C to +45°C), Occupants (1 to 15), and Active_Computers (0 to 15).
- **Output (Target)**: Optimal_HVAC_Temp.



The Algorithm: Decision Tree Regressor

I utilized a DecisionTreeRegressor (with max_depth=7). The algorithm builds a hierarchical tree of rules. For example:

1. *Is the outside temperature below 18°C?* -> **Switch to Winter Heating Protocol.**
2. *Are there more than 5 PCs active?* -> **Lower the heater output by 1°C to save energy.**

This algorithm is ideal because climate control logic is highly non-linear and conditional.

4. Key Decisions Made

Throughout the development process, several critical architectural decisions were made:

1. **Choosing Decision Trees over Linear Regression**: A simple linear model would fail here because the logic inverses based on the season. In winter, extra people mean we need *less* mechanical heat. In summer, extra people mean we need *more* mechanical cooling. Decision Trees handle this "split logic" perfectly.
2. **Implementing Safety Guardrails**: I hard-coded absolute limits (e.g., target_temp = max(16.0, target_temp)). Even if the AI calculates that a room has enough people to stay warm at -45°C outside, the system will never drop the thermostat below 16°C to prevent building infrastructure (like water pipes) from freezing.
3. **Humanized User Interface**: Instead of just outputting a raw integer, I designed the terminal to act as an "Eco-Assistant." It explains *why* it made a decision and displays the estimated Carbon savings.

5. Challenges Faced

1. **Handling Extreme Edge Cases:** Initially, when simulating severe winter temperatures (e.g., -45°C), the model would suggest dangerously low indoor temperatures because it heavily weighted the "free heat" from occupants. I had to refine the data generation loop to enforce safety floors (minimum 16°C) and ceilings.
2. **Balancing the Weights:** Determining exactly how much temperature offset a single human or PC should provide required iterative testing. I settled on an offset of 0.3°C per person and 0.2°C per PC, which created realistic and safe model predictions.
3. **UX in the Terminal:** Making a standard Python console script look like a professional dashboard required careful string formatting, ASCII art, and the integration of the `time.sleep()` module to simulate AI processing latency.

6. Detailed Output Section (System Action)

Below are two sample interactions demonstrating the system's dual-mode logic.

Scenario A: Winter Protocol (High Occupancy)

```
--- 🌱 Welcome back,. I'm your Smart Climate hvac Assistant. ---  
I've analyzed 500+ environmental scenarios to help you save energy today.  
  
--- Current Room Status ---  
What's the temperature outside? (°C): -10  
How many people are in the room right now?: 2  
How many computers/laptops are running?: 1  
  
[AI] is Thinking... analyzing heat signatures...  
  
=====
```

 **RECOMMENDED SETTING: 21°C**

```
=====
```

 It's freezing cold out there! I've increased the intensity of the heater to keep you cozy.
 Sustainability Tip: This temperature setting saves approx. 0.15kg of CO2 today.

```
=====
```

Would you like to check another room? (y/n): y

☀️ Scenario B: Summer Protocol (Low Occupancy)

```
--- Current Room Status ---  
What's the temperature outside? (°C): 45  
How many people are in the room right now?: 12  
How many computers/laptops are running?: 10  
  
[AI] is Thinking... analyzing heat signatures...  
  
=====
```

 **RECOMMENDED SETTING: 18°C**

```
=====
```

 Severe heatwave detected. I'm prioritizing cooling to prevent fatigue. Stay hydrated!

 Note: With 22 heat sources in the room, I've lowered the AC intensity to recycle the natural warmth already present. Smart choice!

 Sustainability Tip: This temperature setting saves approx. 0.3kg of CO2 today.

```
=====
```

Would you like to check another room? (y/n): █

7. What I Learned

This project was a profound learning experience that tied theoretical computer science concepts to real-world applications:

- **Digital Empathy:** I learned that code should not just compute; it should communicate. Adding explanations for AI decisions builds trust with the user.
- **Model Constraints:** I learned the importance of "guardrails" in AI. Machine Learning models will optimize for mathematical efficiency, but engineers must enforce human safety constraints.
- **The Power of Synthetic Data:** I learned how to use numpy to generate realistic data distributions when real-world data is inaccessible, allowing for robust model training.

8. Conclusion

The Eco-Logic system successfully demonstrates how baseline Machine Learning models can revolutionize legacy hardware systems. By utilizing a Decision Tree Regressor to dynamically adjust for internal heat loads, this project proves that intelligent software is just as critical as efficient hardware in the fight for global energy sustainability.

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